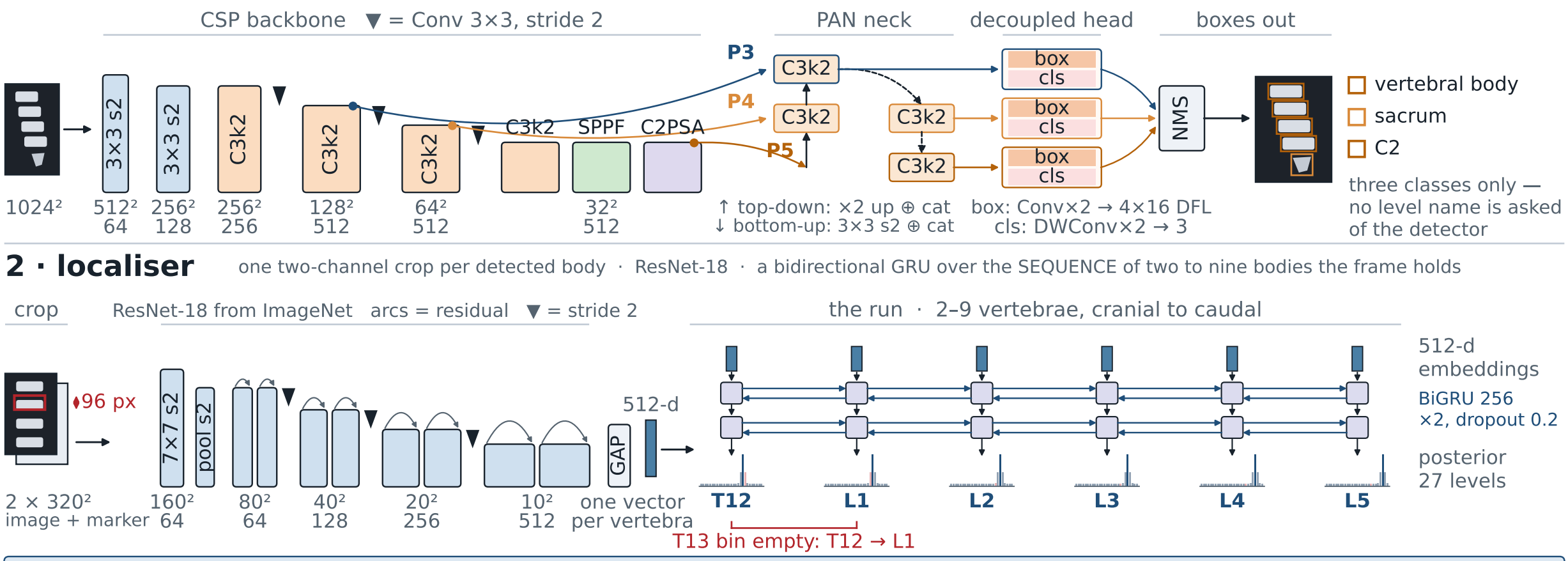
1 · detector
YOLO11m from COCO · 20.1 M parameters · one exposure, 1024 px square · three classes only: vertebral body, sacrum, C2
CSP backbone ▼ = Conv 3×3, stride 2
PAN neck
decoupled head
boxes out
3×3 s2
3×3 s2
C3k2
C3k2
C3k2
C3k2
SPPF
C2PSA
P3
P4
P5
C3k2
C3k2
C3k2
C3k2
box
cls
box
cls
box
cls
NMS
vertebral body
sacrum
C2
1024²
512² 64
256² 128
256² 256
128² 512
64² 512
32² 512
↑ top-down: ×2 up ⊕ cat
↓ bottom-up: 3×3 s2 ⊕ cat
box: Conv×2 → 4×16 DFL
cls: DWConv×2 → 3
three classes only — no level name is asked of the detector
2 · localiser
one two-channel crop per detected body · ResNet-18 · a bidirectional GRU over the SEQUENCE of two to nine bodies the frame holds
crop
ResNet-18 from ImageNet arcs = residual ▼ = stride 2
the run · 2–9 vertebrae, cranial to caudal
96 px
7×7 s2
pool s2
GAP
512-d
2 × 320²
image + marker
160² 64
80² 64
40² 128
20² 256
10² 512
one vector per vertebra
T12
L1
L2
L3
L4
L5
T13 bin empty: T12 → L1
512-d embeddings
BiGRU 256 ×2, dropout 0.2
posterior 27 levels
the recurrence runs between bodies, not inside a crop: a per-crop classifier has no horizontal edges at all.